Euclid's Elements — Book XI
Proposition 3

Formal Reconstruction — Technical Documentation

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CGJteamLab

If two planes cut one another, then their common section is a straight line.

1 Introduction

Proposition XI.3 states that if two distinct planes meet, then their common section is a straight line. In Euclid's sequence this is the natural continuation of XI.2: once planes containing spatial configurations have been introduced, one must also know how two such planes intersect.

At first sight the proposition seems elementary. In fact it is one of the first explicitly dimension-sensitive steps of Book XI. In genuine three-dimensional incidence geometry, two distinct planes with a common point intersect in a line. In higher-dimensional ambient settings this need not be true: two distinct 2-planes may meet in a single point only. Thus XI.3 is not merely a convenient diagrammatic fact. It is a structural property of the three-dimensional spatial layer.

The production file is

```
CGJteamLab/Proposition11_3.lean
```

and the decisive reusable theorem is

```
hilbert_plane_intersection_line
```

from

```
CGJteamLab/Hilbert3DInterface.lean
```

2 Euclid's Statement

Euclid states:

If two planes cut one another, then their common section is a straight line.

The key phrase is "common section". In modern incidence language this means the full intersection of the two planes, not merely the existence of one line lying in both.

The present formal reconstruction therefore states XI.3 extensionally: it returns a line l contained in both planes and proves that a point belongs to both planes if and only if it lies on l.

3 Classical Configuration

Let π and ρ be two distinct planes having a common point A.

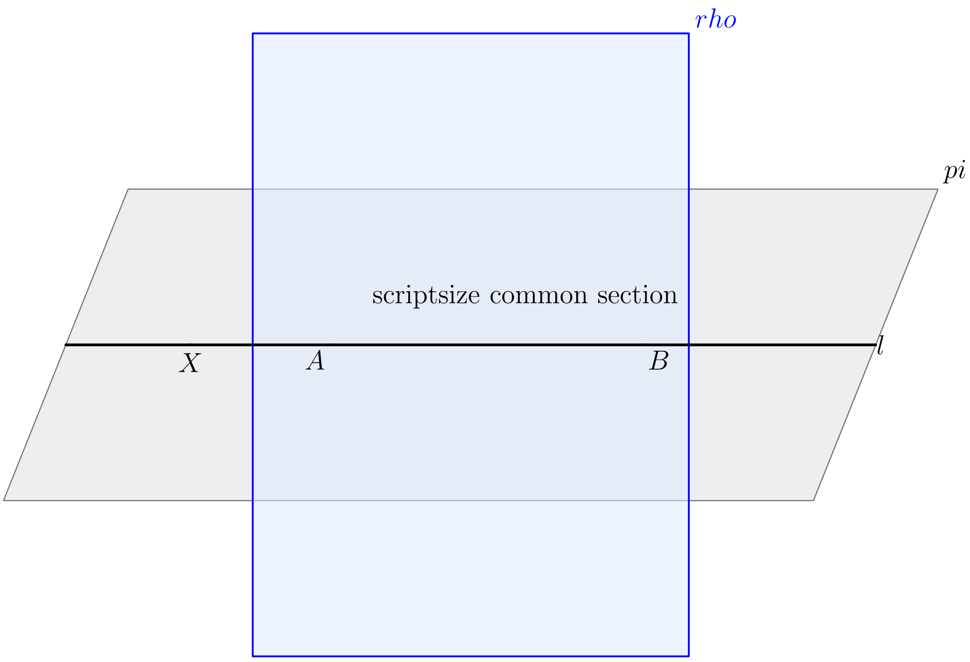
The intended conclusion is that there exists a line l such that

$$A \in l, \qquad l \subset \pi, \qquad l \subset \rho,$$

and indeed that the whole common section satisfies

$$\pi \cap \rho = l$$

in the extensional incidence sense.



scriptsize every common point lies on l

Figure 1: The incidence configuration of Proposition XI.3. The two distinct planes π and ρ have the common point A. Their common section is the black line l. Any point belonging simultaneously to π and ρ lies on this same line.

The important point is that the line is not background decoration. It is the entire common section to be proved.

4 Euclid's Classical Proof

Euclid argues by choosing another common point of the two planes in addition to the given one, and then taking the straight line through the two common points. He then reasons that any point common to both planes must lie on this same straight line, so the common section is a straight line.

At the historical level the proof is short and intuitive. But once one asks what is required for a formal derivation, two hidden obligations appear:

1. from one common point A, one must obtain a second common point B of the two distinct planes;
2. one must prove not only that some line lies in both planes, but that every common point lies on that line.

These are exactly the obligations discharged by the Hilbert 3D interface.

5 Why XI.3 Is Dimension-Sensitive

XI.3 is the first place in Book XI where the three-dimensional nature of the ambient setting becomes mathematically decisive.

In ordinary 3-space, two distinct planes sharing one point must share a whole line. But in a higher-dimensional ambient geometry two distinct 2-planes can intersect in a single point only. Thus the implication

$$A \in \pi \cap \rho, \quad \pi \neq \rho \Longrightarrow \exists l,\ \pi \cap \rho = l$$

is not dimension-free.

In the present project this fact is not left to geometric intuition. It is encoded in the spatial incidence architecture through the principle that two distinct planes with one common point admit a second common point, from which the intersection line is then constructed.

This is the first Book XI proposition whose formal meaning strongly depends on our decision to work in synthetic 3-space rather than in an arbitrary ambient dimension.

6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_3
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (pi rho : S.Plane)
    (hneq : Ne pi rho)
    (A : Geo.Point)
```

```
    (hApi : S.OnPlane A pi)
    (hArho : S.OnPlane A rho) :
    exists l : Geo.Line,
      H.OnLine A l /\
      HilbertLineInPlane Geo l pi /\
      HilbertLineInPlane Geo l rho /\
      forall X : Geo.Point,
        (S.OnPlane X pi /\ S.OnPlane X rho) <->
        H.OnLine X l := by

  exact
    hilbert_plane_intersection_line
      (Geo := Geo)
      pi rho hneq
      A hApi hArho
```

This is a deliberately faithful formalization of the phrase "their common section is a straight line". The theorem does not stop at

$$\exists l,\ l \subset \pi,\ l \subset \rho.$$

It also includes the extensional characterization

$$X \in \pi \cap \rho \Longleftrightarrow X \in l.$$

7 The Stronger Intersection Characterization

The theorem used in the proof is not proposition-local infrastructure. It is the general spatial theorem

```
theorem hilbert_plane_intersection_line
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HilbertSpaceIncidence Geo]
    (pi rho : S.Plane)
    (hneq : Ne pi rho)
    (A : Geo.Point)
    (hApi : S.OnPlane A pi)
    (hArho : S.OnPlane A rho) :
    exists l : Geo.Line,
      H.OnLine A l /\
      HilbertLineInPlane Geo l pi /\
      HilbertLineInPlane Geo l rho /\
      forall X : Geo.Point,
        (S.OnPlane X pi /\ S.OnPlane X rho) <->
        H.OnLine X l
```

Thus XI.3 is a thin Euclid wrapper over a reusable theorem whose conclusion already has exactly the right extensional strength.

8 How the Hilbert Theorem Works

The proof of `hilbert_plane_intersection_line` reveals the real incidence structure hidden behind Euclid's short argument.

Starting from the common point A, the theorem first uses the spatial incidence clause

```
plane_second_common_point
```

to obtain a second common point B such that

$$B \neq A, \qquad B \in \pi, \qquad B \in \rho.$$

Then the ordinary two-point line theorem produces the line l through A and B. Since both points lie in π, Hilbert I.6 implies

$$l \subset \pi.$$

The same argument gives

$$l \subset \rho.$$

The remaining task is extensionality. Suppose X lies in both planes. If X were not on l, then A, B, X would be noncollinear, so by plane uniqueness the plane through A, B, X would have to be both π and ρ. This would contradict $\pi \neq \rho$. Therefore every common point X lies on l.

Conversely, if $X \in l$, then the two line-in-plane containments immediately imply $X \in \pi$ and $X \in \rho$.

9 Formal Dependency Chain

The real formal dependency pattern is therefore

$$
\begin{gathered}
\pi \neq \rho, \quad A \in \pi, \quad A \in \rho \\
\Downarrow \\
\text{second common point } B \\
\Downarrow \\
A \neq B, \quad A, B \in \pi, \quad A, B \in \rho \\
\Downarrow \\
\text{line } l = AB \\
\Downarrow \\
\text{Hilbert I.6} \\
\Downarrow \\
l \subset \pi, \quad l \subset \rho \\
\Downarrow \\
\text{if } X \in \pi \cap \rho \text{ and } X \notin l, \text{ then } A, B, X \text{ noncollinear} \\
\Downarrow \\
\text{Hilbert I.5} \\
\Downarrow \\
\pi = \rho, \text{ contradiction} \\
\Downarrow \\
X \in l \\
\Downarrow \\
\pi \cap \rho = l.
\end{gathered}
$$

This is the real geometric content behind the short public theorem.

10 Spatial Architecture

The theorem uses only the incidence-side spatial structure:

```
HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
```

It does not require

```
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
PlaneGeo
```

as theorem parameters.

Thus XI.3 still lies fully inside the Group I incidence layer. What makes it subtle is not order or congruence, but the need to express a genuinely three-dimensional intersection theorem at the incidence level.

11 Ambient Space and the Absence of PlaneGeo

No induced planar geometry is created in the proof.

All objects remain ambient:

```
pi rho : S.Plane
A X    : Geo.Point
l      : Geo.Line
```

and all relations are ambient incidence relations. The theorem does not enter a slice PlaneGeo(π) or PlaneGeo(ρ) because the result is not a planar theorem inside one chosen plane. It is a statement about the intersection pattern of two ambient planes.

12 Wyler-Style Flat Reconstruction

XI.3 is naturally a meet theorem. Once $\pi \neq \rho$ and a common point are known, the flat layer produces a line l whose carrier is exactly the set-theoretic intersection of the two plane carriers:

$$\mathrm{carrier}(\pi) \cap \mathrm{carrier}(\rho) = \mathrm{carrier}(l).$$

This identity is stronger than merely returning one line lying in both planes: it says that every common point is on the same line and that every point of the line is common to both planes.

The proposition is therefore recovered by a single extensional passage:

```
pi != rho,  A in pi cap rho
        |
        v
Meet(pi,rho) = carrier(l)
        |
        v
X in pi and X in rho  <->  X in l
```

In the Wyler-style vocabulary the meet identity is the normal form of XI.3. It also records why the proposition is specifically three-dimensional: in higher rank, the meet of two codimension-one flats need not have the same rank-one form without additional rank information.

13 Hidden Formal Data

Several facts that are visually obvious in a traditional diagram must be present as proof data in Lean.

13.1 The planes must be distinct

The hypothesis

```
hneq : Ne pi rho
```

is essential. If $\pi = \rho$, then the common section is the whole plane, not a line.

13.2 The common point must be explicit

The theorem takes a witness

```
A : Geo.Point
hApi : S.OnPlane A pi
hArho : S.OnPlane A rho
```

rather than a primitive relation saying merely that the planes intersect.

13.3 The second common point is nontrivial

Euclid's picture encourages one to imagine the common line immediately. Lean must first prove that a second common point exists. This is the hidden three-dimensional step.

13.4 The line must capture all common points

It is not enough to produce one line lying in both planes. The formal conclusion contains the equivalence

```
(S.OnPlane X pi /\ S.OnPlane X rho) <-> H.OnLine X l
```

which states that the line is exactly the whole common section.

14 Relationship with XI.1 and XI.2

Classically XI.3 depends on the coplanarity infrastructure established in XI.1–XI.2.

In the formal development the immediate public theorem does not call `euclid_proposition_11_1` or `euclid_proposition_11_2`. Instead, it uses the stronger general theorem `hilbert_plane_intersection_li` whose proof internally consumes the corresponding Hilbert incidence principles.

The conceptual relation is nevertheless clear:

- XI.1 supplies the line-in-plane closure pattern;
- XI.2 supplies the idea that noncollinear or intersecting data determine a plane;
- XI.3 turns from plane determination to plane intersection.

Thus the classical sequence XI.1–XI.3 is reflected formally in the spatial incidence layer, even though later public theorems may call stronger reusable interface results directly.

15 Relationship with Later Book XI Propositions

XI.3 is a major infrastructure proposition for the remainder of Book XI.

Whenever a later argument introduces an auxiliary plane and then needs its section with another plane, XI.3 is the conceptual bridge. This becomes especially important in propositions such as XI.13, where the proof passes to the auxiliary plane determined by two lines and then studies its intersection with a reference plane.

In the Lean development, later propositions typically consume the reusable theorem `hilbert_plane_intersecti` rather than the Euclid wrapper `euclid_proposition_11_3`. The mathematical content, however, is exactly XI.3.

16 Classical DAG versus Formal DAG

The difference between the historical and formal organizations is again instructive.

Euclid.

$$\begin{gathered}
\pi, \rho \text{ meet} \\
\Downarrow \\
\text{take another common point} \\
\Downarrow \\
\text{join the two common points} \\
\Downarrow \\
\text{any further common point lies on the same join} \\
\Downarrow \\
\text{XI.3.}
\end{gathered}$$

Lean / Hilbert.

$$\begin{gathered}
\pi \neq \rho, \quad A \in \pi \cap \rho \\
\Downarrow \\
\texttt{plane_second_common_point} \\
\Downarrow \\
B \neq A, \quad B \in \pi \cap \rho \\
\Downarrow \\
\texttt{line_through}\ A, B \\
\Downarrow \\
\texttt{line_in_plane}\ \text{twice} \\
\Downarrow \\
l \subset \pi, \quad l \subset \rho \\
\Downarrow \\
\texttt{plane_unique} \\
\Downarrow \\
\forall X,\ X \in \pi \cap \rho \leftrightarrow X \in l \\
\Downarrow \\
\texttt{euclid_proposition_11_3}.
\end{gathered}$$

The formal DAG is longer because Euclid's phrase "common section" has been made fully extensional.

17 Structural Lesson

XI.3 illustrates a basic principle of spatial formalization: once several planes are present, their intersections cannot remain visual background. They must be represented by explicit proof objects and characterized extensionally.

This proposition also shows that "working synthetically" does not mean leaving dimensional constraints implicit. On the contrary, the synthetic reconstruction isolates exactly the point where 3-dimensionality matters.

The theorem is still incidence-only, but it is no longer merely elementary bookkeeping. It identifies one of the structural laws that make Book XI possible as a theory of solid geometry rather than a collection of planar pictures drawn in space.

18 Audit Status

This documentation was checked against the current production declaration `Geometry.euclid_proposition_11_` and the reusable plane intersection theorem on which it is based. The extensional description of the common section and the classical/formal dependency comparison in this note reflect the current production architecture.

19 Conclusion

Proposition XI.3 states that two distinct planes with a common point have a common section which is a straight line.

In the present reconstruction this is expressed in its strongest natural incidence form:

$$\pi \neq \rho, \quad A \in \pi \cap \rho \Longrightarrow \exists l,\ A \in l,\ l \subset \pi,\ l \subset \rho,$$

together with the extensional characterization

$$X \in \pi \cap \rho \Longleftrightarrow X \in l.$$

The proof is short at the public level, but the underlying helper theorem reveals the real 3D structure: from one common point one obtains a second, from two common points one constructs the section line, and plane uniqueness forces every further common point onto that same line.

Thus XI.3 is not just a visual fact about two drawn sheets. It is one of the first genuinely structural theorems of synthetic 3-space in the Book XI development.